Modeling semantic business trajectories of territories for multidisciplinary studies through controlled vocabularies

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***Abstract*— Environmental changes influence society and this impact extends to businesses across several sectors. The risks linked with environmental changes have both advantages and disadvantages for companies. Therefore, business analysts must understand how their companies should deal with the opportunities and challenges that environmental changes present. On the other hand, environmental analysts working on different aspects (e.g. atmospheric sciences, ecology, geosciences, social sciences, etc.) of an environment for human health and well-being are also interested in studying the impact of businesses (e.g. the launch of a factory) on society. Though, business and environmental analysts are from different educational backgrounds. They work on different vocabularies and data formats for analyzing data. These differences create semantic heterogeneities to understand key concepts from data. This makes understanding cross-domain data a complex and time-consuming process for them, eventually creating a knowledge barrier. To address this knowledge barrier, this work presents a proof-of-concept modeling of business trajectories using online news data. The proposed method consists of five stages: a) news data preprocessing, and topic model training for identifying the relevant thematic concepts related to business trajectories using the historical dataset of news articles, b) semantic enrichment of thematic concepts using the controlled vocabularies, c) processing latest news articles using the trained topic model to acquire the most similar thematic concepts, and obtaining the location and temporal entities using the Named-Entity Recognition model (NER) process, d) constructing a knowledge graph using the collected thematic, spatial and temporal entities, along with enrichment of the relevant environmental data, and e) visualizing the semantic trajectories from knowledge graphs for understanding insights in data. This article presents the potential for retrieving the same trajectory data using different business and environmental viewpoints for multidisciplinary data analysis.**

Keywords—Business trajectories, Domain-agnostic, Linked data, NER, Topic modeling

I. Introduction

Environmental changes in the geographic territories present both benefits and disadvantages to local businesses [1]. Business analysts must shape their company strategies such as company movements, market shares, the launch of products and services, legal issues, and technological changes in conjunction with environmental data [1, 2]. Environmental changes influence market reactions and present business opportunities for companies. For instance, the decrease in temperature in the winter, a lack of sunlight, and dry air negatively affect the ability of inhabitants to protect themselves from respiratory infections such as coronavirus [3]. This will present many business opportunities related to the production of antibacterial supplies and the offering of cleaning services, offers on online food delivery, and telemedicine facilities. Another example would be, that the increase in the population of a geographic territory will increase consumer demands, which requires an increased scale of production of goods by local industries.

News articles present a promising way to capture up-to-date information about companies' current and future movements [4]. However, this information is incomplete if it is not mapped with the environmental data associated with their geographic territories of origin. The environmental data will provide the context for business news that can be used by business analysts to support their decisions. On the contrary, the environmental analysts working to monitor and evaluate the health conditions of our environment also need business data. For example, they need data on new industries that were set up in the territory and that can lead to the depletion of natural resources. In addition, how can new market products or services cause air pollution, soil pollution, and water pollution or impact global warming or climatic changes?

But, the requirement for one-time data acquisition from the news articles related to the business and environmental domain does not end here. The business news data evolves over time. For instance, it is important to understand how the particular business opportunity (i.e. represented as a collection of concepts) in a geographic territory has triggered other business opportunities or the same opportunity in other territories over time. At the same time, this business opportunity can also present environmental challenges. To keep track of the evolution of business opportunities in the market, the acquisition of continuous data from news articles on a daily basis is required [4]. For this, the idea of constructing business trajectories is formulated. The business trajectory is a collection of traces (i.e. business opportunities) generated by an evolving business topic in geographic territories represented using chronologically ordered date-timestamps [5]. Adding relevant context (i.e. additional business and environmental information) to this business trajectory that can be used to characterize the situation of the business topic leads to the construction of the semantic trajectory.

The rest of the paper is organized as follows: In Section 2, the background is discussed. In Section 3, the proposed method is presented. In Section 4, the presented method is discussed. Finally, in Section 5, the article concludes by system limitations and potential future research directions.

II. Background

The term "trajectory" signifies a path under the influence of a force, which describes the evolution of an entity [5]. This entity can be a person, object, or concept. Formally, a trajectory T is the graph of a continuous mapping from $I \subseteq R$ to R^2 (the two-dimensional plane) [5].

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$$I \subseteq R \rightarrow R^2 : t \rightarrow \propto (t) = \left(\propto_x (t), \propto_y (t)\right)$$

$$T = \left\{\left(\propto_x (t), \propto_y (t), t\right) | \, t \in I\right\} \subset R^2 \times R$$

The trajectory connected to a data sample should contain the sample points, i.e. for all points (x_i, y_i, t_i) in the sample we have $(x_i, y_i, t_i) = (\propto_x (t_i), \left(\propto_y (t_i), t_i\right)$. If the sample points are ordered in time, i.e. if $i < j$ then $t_i < t_j$, then this order is preserved in the trajectory [5]. In addition, given a data sample, there is an infinite number of trajectories linked to that data sample. The trajectory is by no means unique [5]. Previously, the notion of a trajectory was used to describe the movement of an entity collected through a Global Positioning System (GPS), Global System for Mobile Communications (GSM), Radio-frequency identification (RFID), and sensor techniques [5-7]. But, in this study, the idea of a trajectory is used specifically for a collection of different business and environmental topics that evolves over time and consists of different concepts. Concepts are referred to as abstract ideas and are regarded as the fundamental building blocks which form topics. For instance, the concept "construction" defines the idea related to a business domain involving the structuring of new buildings, etc. Moreover, the concept of "climate change" is centered on the shifts in temperature and weather patterns. The formerly mentioned concept is an example of a business concept. This is because the construction of a new building is treated as a potential business opportunity for different people or companies to find jobs related to cleaning, site management, interior design, etc. respectively to the type of building. Whereas, the latter concept is an environmental concept that defines the parameters of the environment of a geographic region.

The concepts must undergo the process of semantic annotation to support multidisciplinary studies [8]. Semantic annotation of data facilitates identifying matching data elements existing in multiple sources and thus supports data integration [10]. In our case, it will ensure that the concepts can be explored by people from different backgrounds in business and environmental science. There exist many studies in the literature related to the techniques as well as the knowledge sources and ontologies that can be used for the semantic annotation of data. Milosevic [9] annotated text data using WordNet, MetaMap, DBPedia, and thesauri characterized as SKOS (Simple Knowledge Organization System). In addition, the provenance data (i.e. information related to the history of an entity to verify its quality) is added using PROV-O ontology. Dugas et al. [10] proposed to construct semantic annotations of medical data using the large-scale public portal of medical data models (MDM) (i.e. an open-access repository). An open XML-based data standard is used for establishing semantic annotations. Yan et al. [11] used land use data, points of interest data, and road network data to annotate the GPS traces for different applications such as urban planning, fleet tracking, traffic analysis, etc. There are numerous other studies that also exist, and the reader is referred to the detailed survey article of Liao & Zhao [12] for more examples of semantic annotations.

The number of concepts along with their annotations can be numerous and thereby needs some organization. For that, similar concepts are maintained under the same topic. A topic is usually more generic and includes several specialized concepts. For instance, a topic about renewable energy may contain concepts such as power plants, solar power, photovoltaic parks, etc. In the literature, there exists different topic models for discovering and organizing abstract topics automatically as a set of concepts that occur in a set of documents by considering the underlying hidden semantic meaning in a text [13]. Traditional approaches, such as Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF) define a text document as a bag of words (i.e. presence of words along with their frequency) and model each document as a collection of topics [13-19]. One of the major drawbacks of LDA-based topic models is that the semantic relationships among words are not considered and thus fail to accurately represent documents. To overcome this problem, Bidirectional Encoder Representations from Transformers (BERT)-based topic models such as BERTopic [13, 16-19] have demonstrated promising results by considering the underlying semantic context in documents. BERTopic uses the Sentence-BERT (SBERT) framework [19] and transforms sentences into dense vector representations using language models. The vector representations are grouped using the similarity index, and finally, topic representations are extracted using a custom class-based Term Frequency–Inverse Document Frequency (TF-IDF). More details on BERTopic can be found at [13].

Once topics having different concepts are extracted, they are viewed as trajectories because they evolve over time through geographic territories. The topics as trajectories that are related to business and the environment in our case, can cross, diverge or maintain their paths. But to understand the context behind such trajectory sequences, we need to go beyond the thematic data (comprising the domain-related concepts) and acquire more information about these trajectories. Here, comes the idea of annotating the trajectory with a semantic location and timestamp. The semantic location corresponds to the geographical region where the business or environmental trajectory is taking place. Whereas, timestamp maps the temporal information with the trajectory. By coupling the spatio-temporal information, we can find out additional contextual information related to the environment (e.g. temperature, atmospheric CO2, and human population) where the business trajectory is happening. Eventually, we can achieve a trajectory by having the business or the environmental information mapped with location and time. So, we define the semantic trajectory as "a trajectory having the semantic annotations related to its theme, geographical location, and timestamp" [5, 6]. Semantic Web (SW) technologies have been used widely in the literature to model semantic trajectories, since they facilitate the interlinking of data that could enhance a trajectory of raw data (spatio-temporal) points based on semantic information, in a standardized and meaningful way [5-7]. Hence, the collected information about the trajectories is stored as Knowledge Graphs (KGs). A knowledge graph represents a network of real-world entities (e.g. concepts) and made up of three main components which are nodes, edges, and labels [20]. This information is kept in a graph database. The construction of KGs present tremendous applications to enable the extraction of insights by combining siloed data sources and maintaining interconnected relationships to support data-driven and knowledge-intensive semantic search and reasoning [21]. However, exploring different applications of KGs is not the scope of this work.

III. Proposed Method

The proposed method (see Fig. 1) consists of five stages which are; a) news data preprocessing, and topic model

training for identifying the relevant thematic concepts related to business trajectories using the historical dataset of news articles, b) semantic enrichment of thematic concepts using the controlled vocabularies to support multidisciplinary studies, c) processing latest news articles using the trained topic model to acquire the most similar thematic concepts, and obtaining the location and temporal entities using the Named-entity model (NER) process, d) constructing a knowledge graph using the collected thematic, spatial and temporal entities, along with enrichment of the relevant environmental data, and e) visualizing the semantic trajectories from the knowledge graphs for understanding insights in data.

The 1st step is to construct the collection of relevant business concepts of interest, which defines the underlying context. For this, a dataset of news articles (i.e. 20,000 text files) is used which only contains the relevant articles that have been filtered manually. These news articles were crawled from 95 different news sources through their original websites, such as lamontagne.fr, fusacq.com, businessimmo.com, sudouest.fr, actu.fr, ladepeche.fr, lavoixdunord.fr, etc. The publishing time of these news articles spans five years, from 2017 to 2021. The news articles were translated into English using the Deep Translator Python library (i.e. https://pypi.org/project/deep-translator/) and basic preprocessing steps such as stop-word removal, and filtering of special characters, weblinks, email addresses, and punctuations were executed using the Natural Language Toolkit (NLTK) (i.e. https://www.nltk.org/). After preprocessing the dataset, a corpus of around 1,324,859 words was built.

To extract the topics as a collection of concepts from the built text corpus, the application of a topic model is required. There exist many topic models in the literature. BERTopic [13] is chosen because of its efficiency and popularity in the understanding the underlying semantics in text to create dense clusters for easy interpretable topics using transformers and c-TF-IDF technique. The state-of-the-art BERTopic model [13] is trained on the corpus of news articles and topics were extracted (see Fig. 2). Each topic includes a group of concepts. These concepts are visualized by constructing bar charts out of the c-TF-IDF scores for each topic representation. This will help to compare concept representations to each other (see Fig. 3). After training the topic model, it can now be used for predicting the topics (as a set of concepts) of the news articles.

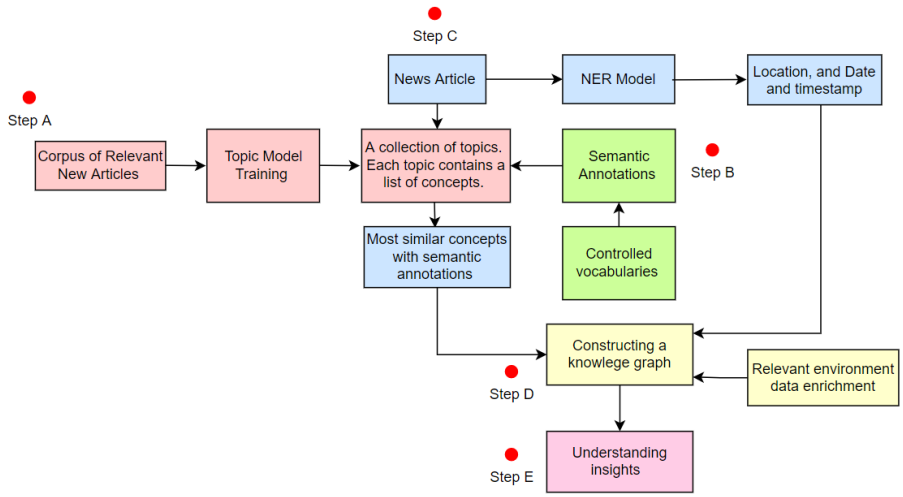

Fig. 1. Proposed method

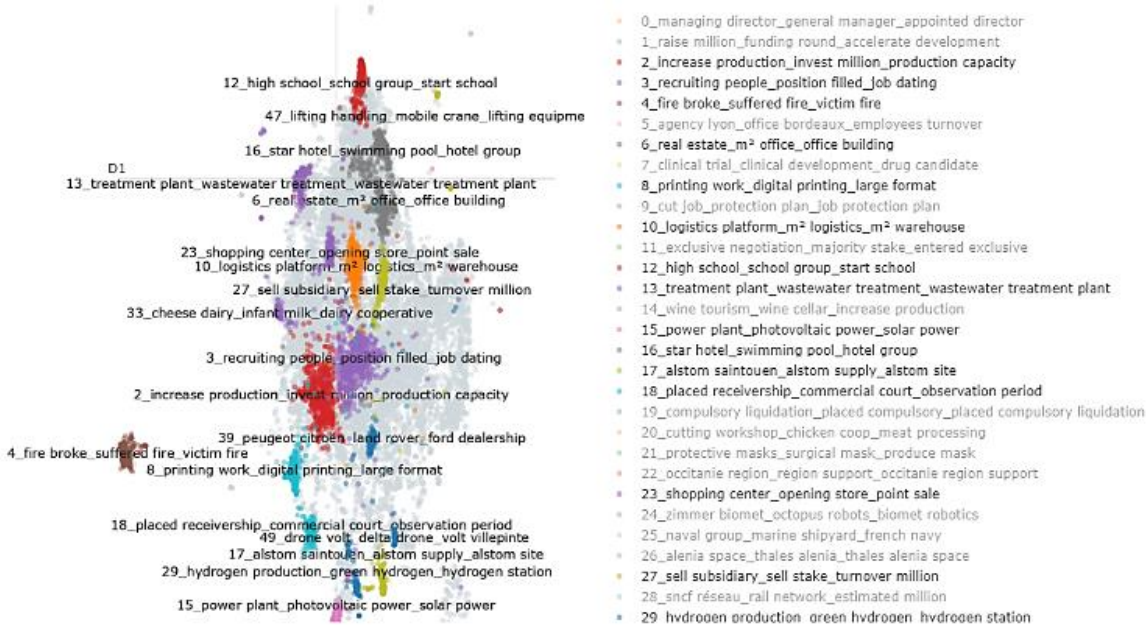

Fig. 2. Different topics extracted from a set of news documents

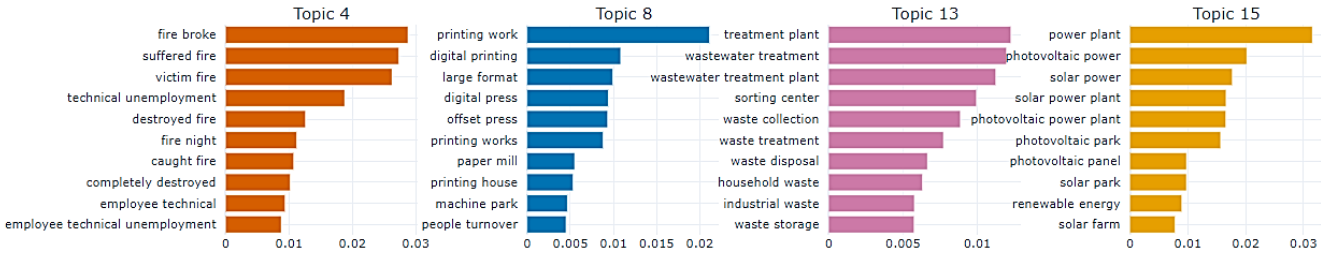

Fig. 3. Concepts' scores

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[ ] similar_topics, similarity = topic_model.find_topics(ArticleTitle&Description, top_n=3); similar_topics
```

Fig. 4. Searching for the most similar topics which can describe the context of a news document

To extract the topics for new articles, we used the built-in function i.e. "find_topics" of the trained topic model [13] to search for topics that corresponds to the content of news articles (see Fig. 4). This function is executed by passing the article`s title and description text (see Fig. 4) as an input. This will enable us to identify the most similar topics (i.e. a set of concepts) automatically. To make the concepts domain-agnostic (i.e. generalized concepts that are interoperable to use for different research communities) to support multidisciplinary studies ideally, all identified topics having concepts which are identified in Step A should be considered. However, to show the proof-of-concept working, only three concepts from different topics are semantically annotated using controlled vocabularies which are specific to the business and environmental domains (see Fig. 5, 6 and 7). The vocabularies (EUROVOC, GEMET and AGROVOC) are chosen after exploring their relevancy in the existing literature [22-24] for business and environmental domains. The concept 'economic support' is annotated using EUROVOC (i.e. the EU's multilingual and multidisciplinary thesaurus) and GEMET (i.e. GEneral Multilingual Environmental Thesaurus). Whereas, the concepts 'population' and 'economic development' are annotated using EUROVOC and AGROVOC (i.e. a multilingual controlled vocabulary). The purpose of annotating with different vocabularies is that, the point-of-entry (i.e. inputing a concept-of-interest to system) to initiate the data searching process cannot be same for business and environmental analysts. Their begin the searching process based on their domain-specific vocabulary. To widen their scope of exploration for concepts which are not present in their domain-specific vocabularies, the process of semantic annotation is achieved here.

The SKOS model is used for annotating the concepts because it provides a standard method for representing knowledge organization systems using the Resource Description Framework (RDF) [25, 26]. Using RDF for encoding concepts will allow them to be passed between different computer applications in an interoperable mode. Moreover, the SKOS model helped us to organize the concepts relating to each other by hierarchical links. For example, the *skos:broader* property for ascending hierarchical link, the *skos:narrower* for downward hierarchical link, and *skos:related* property for associating a neighboring concept with the concept. Furthermore, to establish links between vocabularies, we have used the *skos:exactMatch* property indicating two concepts are equivalent, and the *skos:closeMatch* property specifies that the two concepts are semantically close [25, 26]. After annotating the concepts, NER is used to identify named entities in text. NER is an important component in Natural Language Processing (NLP) systems, and used to find entities such as person, organization, time, location, etc. in text [27]. In our case, we have chosen two named entities, which are geographical locations and date timestamps. There are different Java and Python-based libraries available for executing the NER on text. Some of them are SpaCy, Apache OpenNLP, and TensorFlow [28].

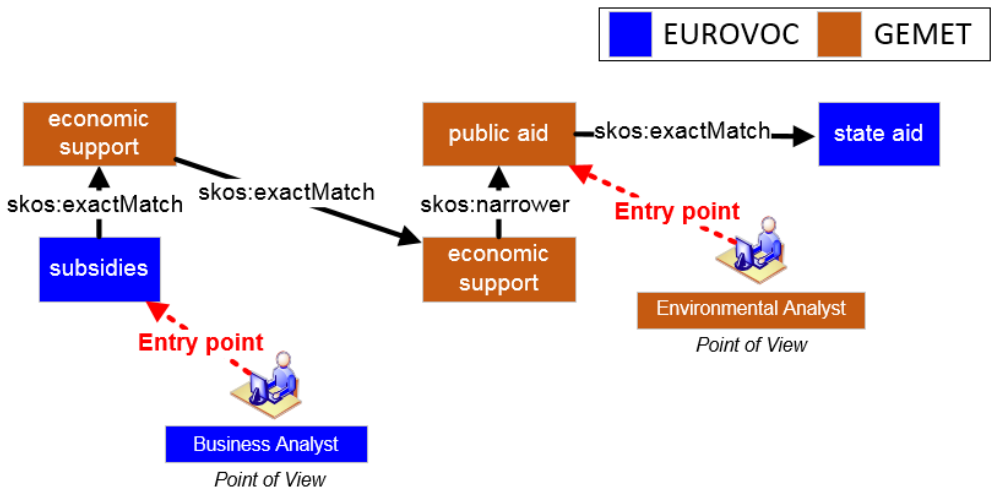

Fig. 5. The concept 'Economic support" is defined to support domain-agnostic search

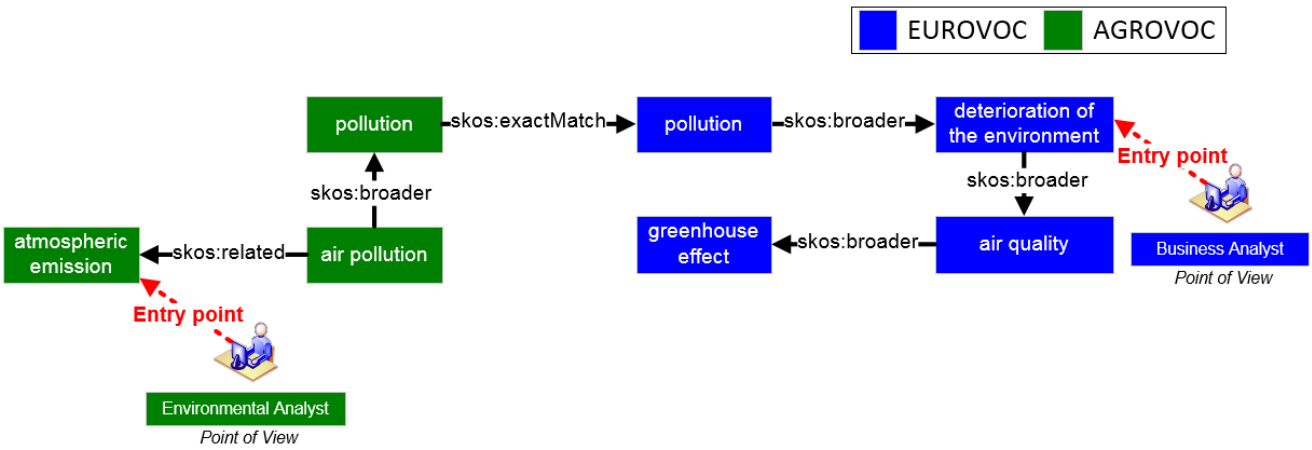

Fig. 6. The concept 'Pollution" is defined to support domain-agnostic search

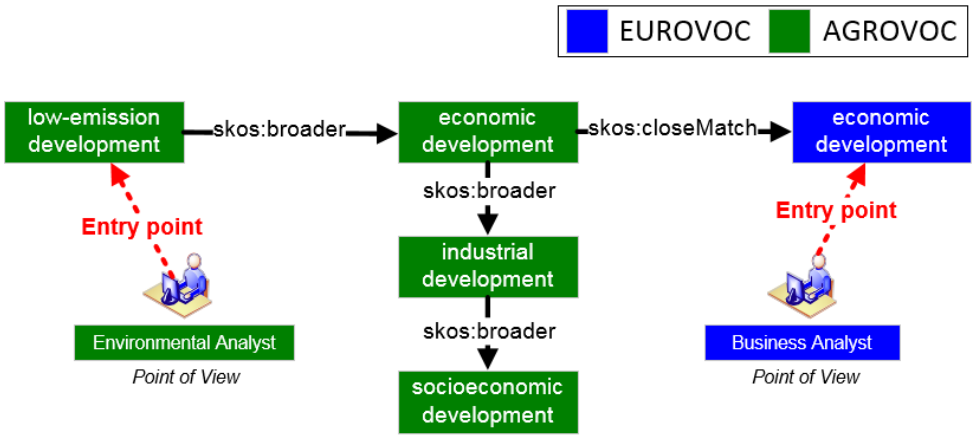

Fig. 7. The concept 'Economic development" is defined to support domain-agnostic search

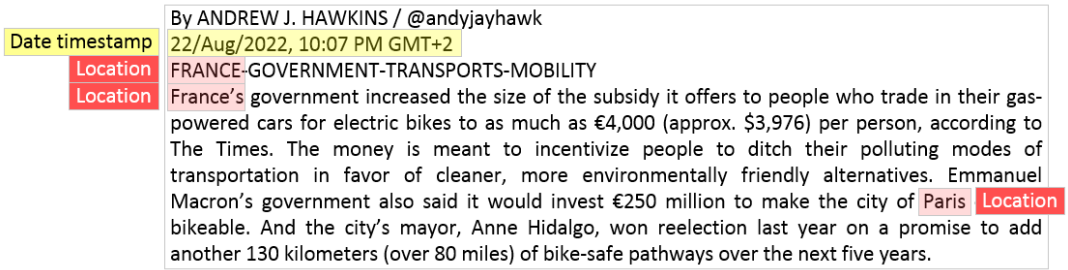

Fig. 8. Named-entity identification

Shelar et al. [28] found that SpaCy gives better and more accurate results as compared to Apache OpenNLP and TensorFlow to identify named entities in the text. Henceforth, SpaCy is used here (see Fig. 8).

After obtaining the topics and the values of spatio-temporal entities through topic modeling and NER, we can now construct the KGs. The purpose of using the KGs is that it helps to model the semantic trajectories (business trajectories in our case) in a standardized and meaningful way using SW technologies. To model the KGs for representing business data along with environmental data for news articles, the fundamental requirements were initially found through the literature review [4, 8]. These requirements are presented in Table 1. The identified requirements were based on the data related to the news articles i.e. their unique identification in the system (article id), title, description text, the URL from where the article can be accessed, their domain (identified as topics using a set of thematic concepts earlier), the date of publishing, and the location entity linked with them that is found using the NER process. To add the environmental data, we used temperature, population, and pollution level readings based on the identified location and date-time entities found in the text of news articles (see Fig. 9). More environmental entities can also be added based on their availability for modeling them with business data for analysis. To model the KGs, relevant state-of-the-art ontologies and vocabularies were used to address the required questions (mentioned in Table 1), which are; 1) Resource Description Framework Schema (RDFS) (https://www.w3.org/TR/rdf-schema/), DCTERMS(https://www.dublincore.org/specifications/dublincore/dcmi-terms/), and DCAT ontology (https://www.w3.org/TR/vocab-dcat-2/). In addition, the KG uses GeoNames (https://www.geonames.org/) for specifying the spatial features and OWL-Time ontology (https://www.w3.org/TR/owl-time/) for describing temporal

concepts. The reason for selecting these ontologies and vocabularies is straightforward since they are compliant with the Open Geospatial Consortium (OGC) providing an open semantic data infrastructure [29]. The creation of KGs opens many possibilities to design different solutions based on a variety of tasks concerning question-answering, recommendation engines, and information retrieval [8, 20, 23]. However, for this article, only the visualization of semantic trajectory data acquired from the KGs is used to understand the evolution of business opportunities concerning the change of temperature across geographical territories. The news articles that correspond to topic 15 (see Fig. 3) that is "solar power plants" is visualized with temperature values of the geographical regions during 2017. As it can be observed (see Fig. 10), over time, which regions are publishing the business opportunities related to "solar power plants"? Also, how these regions are triggering business opportunities in other regions. Moreover, it will help the analysts to analyze the dependence of temperature with respect to regions, which is important to generate business opportunities related to "solar power plants".

IV. Discussion

Many new topics in the form of different concepts appear in news articles over time. However, these concepts are still scattered, i.e., they contain information about business and environmental domains but are problematic to reuse and integrate with the existing ones [4, 8]. It is a significant challenge to retrieve the data related to these emerging concepts and make it possible to reuse them. Achieving this would promote the scientific discovery of data for analysts involved in conducting different multidisciplinary studies.

This work has identified how to fetch evolving topics as a set of concepts from news articles automatically. This is made possible with the use of the state-of-the-art topic model. Later, these concepts are semantically-enriched using controlled vocabularies to support multidisciplinary studies. For example, using the proposed model, business and environmental analysts can retrieve data related to economic support in different ways. The business analyst, interested in studying economic support, identifies subsidies as the main context (i.e. a point of entry for starting the process of searching for the data). However, this point of entry does not necessarily have to be fixed to start the data searching process by the environmental analyst. The data relating to economic support can be explored even if the broader concept of economic support such as 'public aid' or the same concept such as 'subsidies' of another business-controlled vocabulary is entered by the analyst (see Fig. 5). Similarly, the environmental analyst, interested in studying pollution might identify atmospheric emission as the point of entry for starting the data search. But, this concept can also be discoverable through multiple concepts such as 'air quality', 'deterioration of the environment', and 'greenhouse effect' found in the

TABLE I. Requirement Questions with Entities and Linking Properties to Model Knowledge Graphs

No.	Requirement questions	Entity	Used property
1	What is the article id?	ID	hasID
2	What is the title of a news article?	Title	dct:Title
3	What is the description of a news article?	Description	dct:Description
4	What is the domain of a news article?	Domain	dcat:Keyword
5	What is the location entity mentioned in a news article?	Location	gn:Feature
6	Where to access full news article?	URL	dct:DownloadURL
7	What is time entity found in a news article?	Date	dct:Temporal
8	What is the temperature of the geographic region?	Temperature	has
9	What is the population of the geographic region?	Population	has
10	What is the pollution level of geographic region?	Pollution level	has

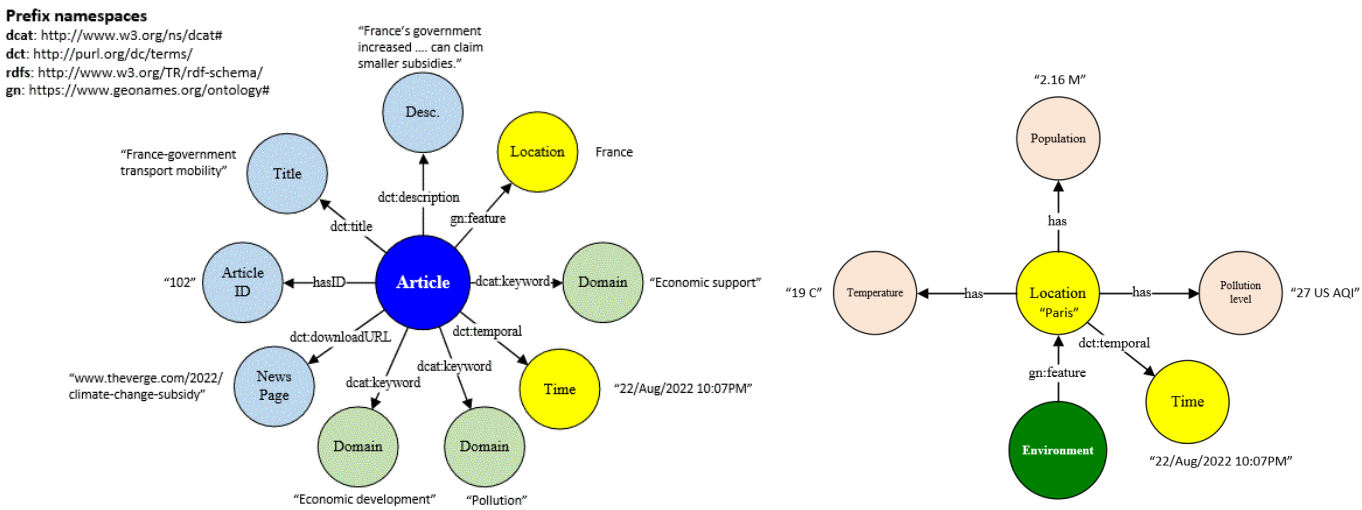

Fig. 9. a) Knowledge graph for business data (left), b) Knowledge graph for environmental data (right)

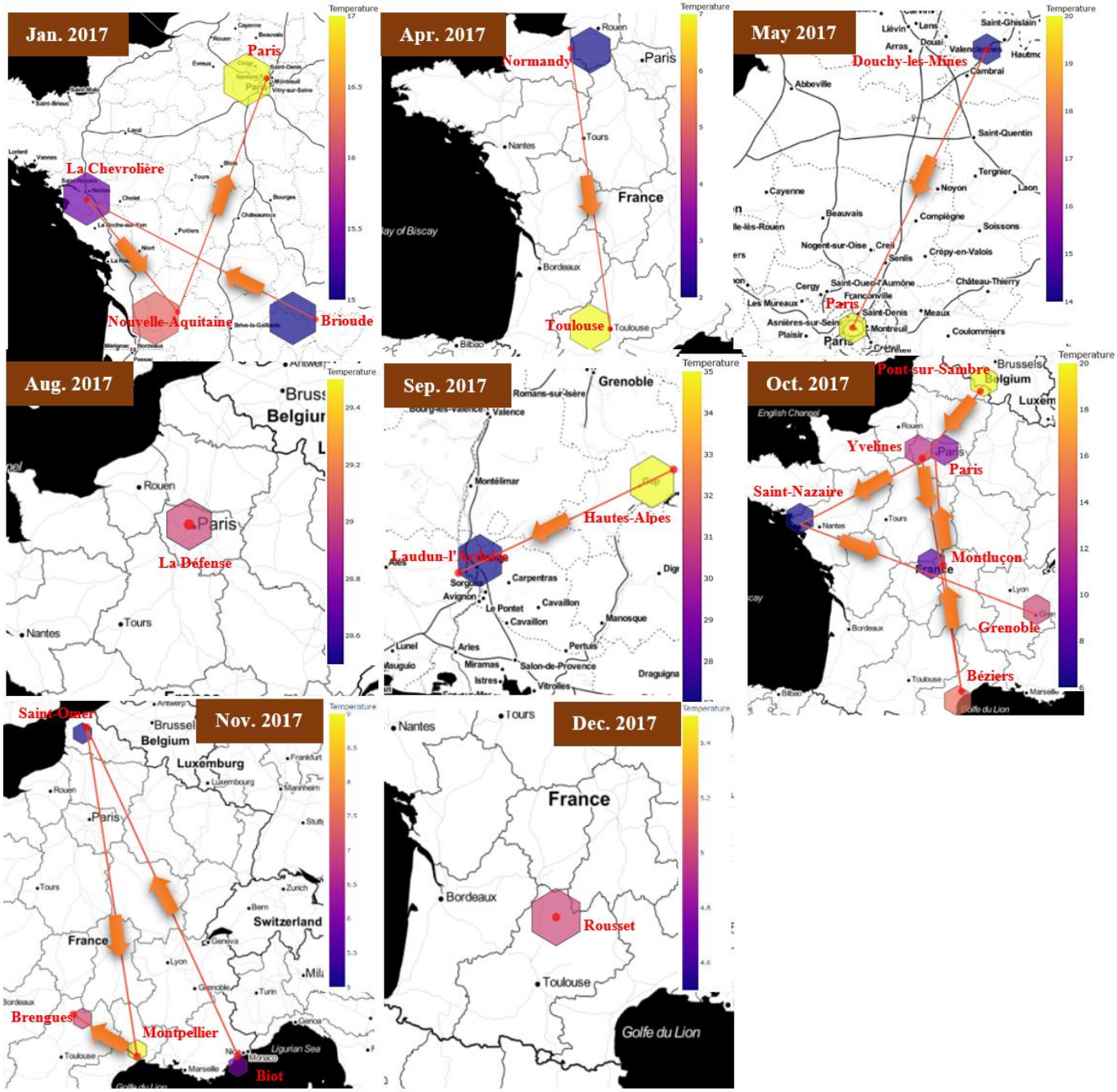

Fig. 10. An example of a semantic trajectory of news articles related to "solar power plants" during the year 2017 that is mapped with average temperature values of months based on their geographical territories.

AGROVOC (i.e. more specific to environmental sciences) (see Fig. 6). Furthermore, the business analyst, interested in studying economic development, explores more data related to its broader concepts, such as 'industrial development' and 'socioeconomic development' through semantic annotations made through the SKOS model (see Fig. 7). After the process of semantic linking of concepts, the latest news articles are processed using the trained topic model that has helped us to obtain the most similar topics containing different concepts. In addition, the NER model has helped us to get the location and temporal entities related to the extracted topics. The identified thematic and spatio-temporal entities are later stored as KGs to model semantic business trajectories integrated with environmental data.

The integrated business data with environmental data that is visualized as semantic trajectories will help us to understand the evolution of hot (i.e. gaining popularity) and cold (i.e. losing popularity) business topics over time across geographical territories concerning environmental variables. For instance, in this work, the trajectory of business opportunities from news articles on "solar power plants" is visualized over regions during 2017. During the trajectory analysis, the point (i.e. a geographical region) where a business trajectory is originating or splitting towards different regions or terminating can be a critical point of interest for analysts. However, this work can be considered as a basis for building systems to carry complex forms of semantic trajectory analysis in the future.

V. Conclusion

The novelty of the proposed model lies in the integration of techniques related to knowledge graphs and NLP to provide an informative and engaging view of business topics spatially, temporally, and contextually over time. The spatial information of a business trajectory will reveal whether a business topic is local or has a particular global political or economic facet based on the geographic territory. Also, temporal information will reveal the business topic's importance and duration. Furthermore, the contextual information of the trajectory will show how different geographic territories come up with the same business opportunities as competitors in the market. This is made possible by modeling KGs in this study, which is based on SW technologies. The application of SW technologies helps us to link corresponding environmental data of geographic territories with business and environmental topics in a standardized and meaningful way. The constructed KGs present a starting point to explore the opportunities to understand semantic business trajectories spanning different geographic territories combined with environmental data for multidisciplinary studies and inferring insights.

The major benefit of the developed KGs is that they enable us to retrieve the same data of semantic trajectories using different viewpoints. This is made possible by the semantic linking and enrichment of concepts using existing controlled vocabularies. However, the semantic linking and enrichment of the concepts are achieved manually. Manual labeling is time-consuming and is not suggested for labeling a large number of concepts in a short time. Existing text-mining methods combined with machine learning techniques can be used for automatically executing semantic enrichment of concepts to support multidisciplinary studies.

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